

Internet of Things

Module 1: Definition

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Internet of Things

Definition:

Considering the wide background and the number of required technologies in IoT, proposing an unambiguous definition of the “IoT” is non-trivial.

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Definition:

Let us start with a very simple definition:

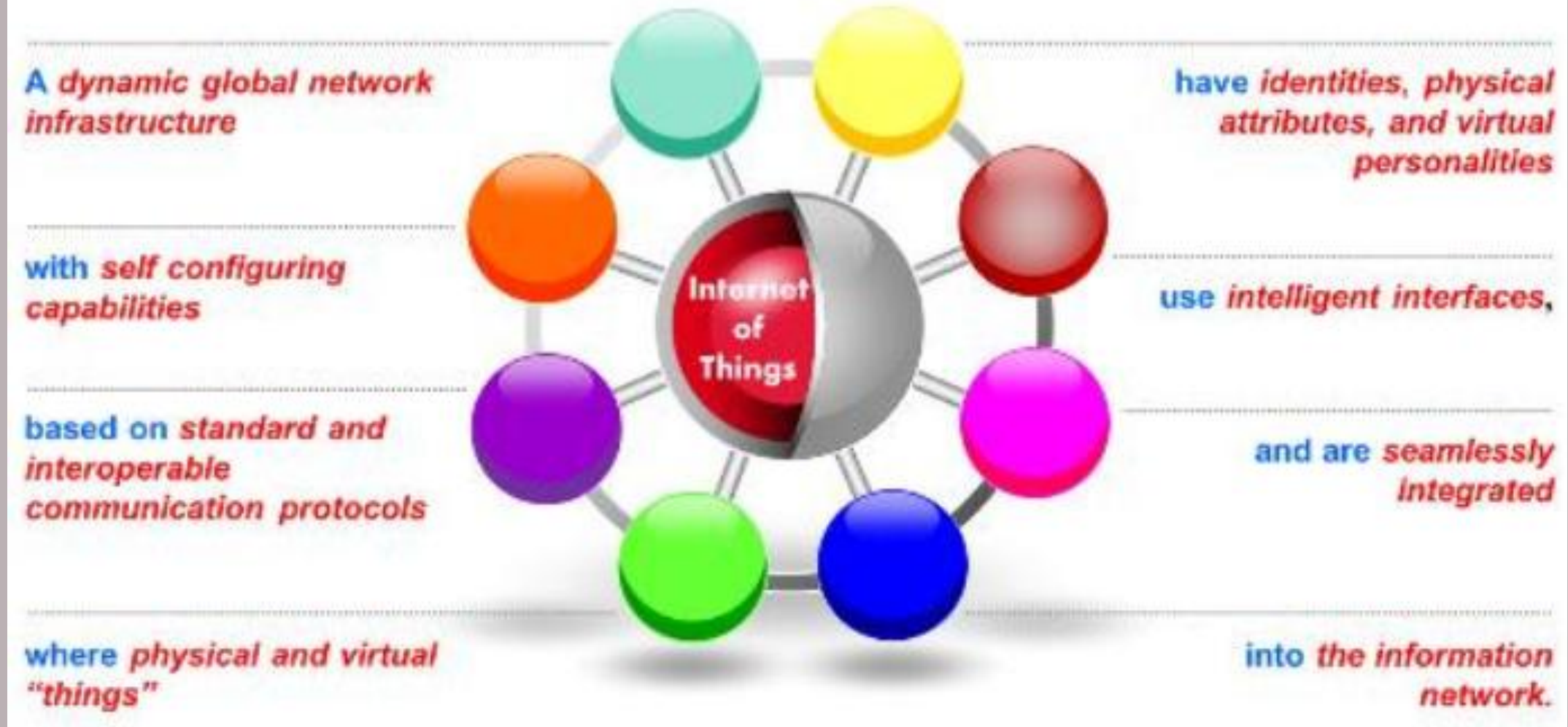
IoT is a network of uniquely identifiable connected *things* (also known as devices, objects, and items) offering intelligent computing services.

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However, a widely accepted definition has been suggested by the ITU-T as

Definition: A **dynamic and self-adapting** global network infrastructure with **self-configuring capabilities** based on **standard and interoperable communication protocols** where physical and virtual "things" have **identities, physical attributes, and virtual personalities** and use **intelligent interfaces**, and are seamlessly **integrated into the information network**, often communicate data associated with users and their environments.

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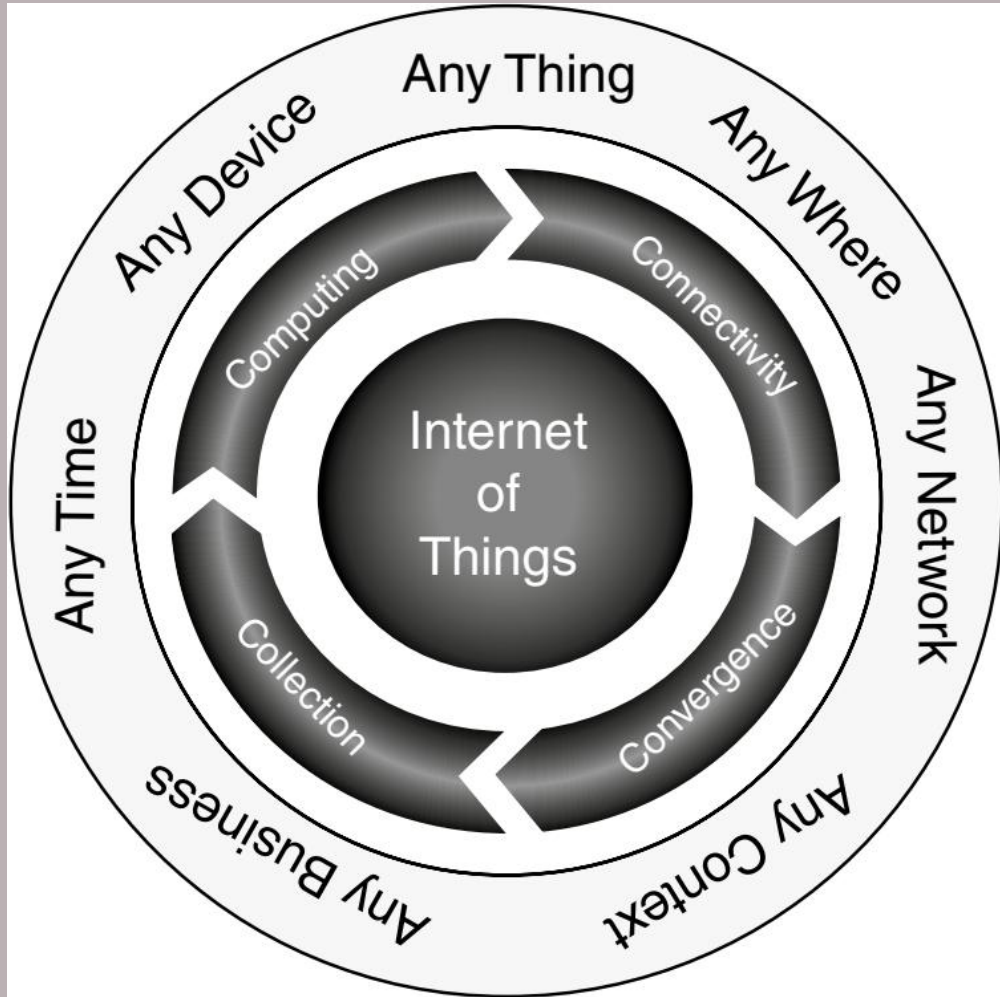


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Definition:

- The fundamental objective of IoT is to provide seamless **ubiquitous (pervasive) connectivity** among things and human beings and machines.

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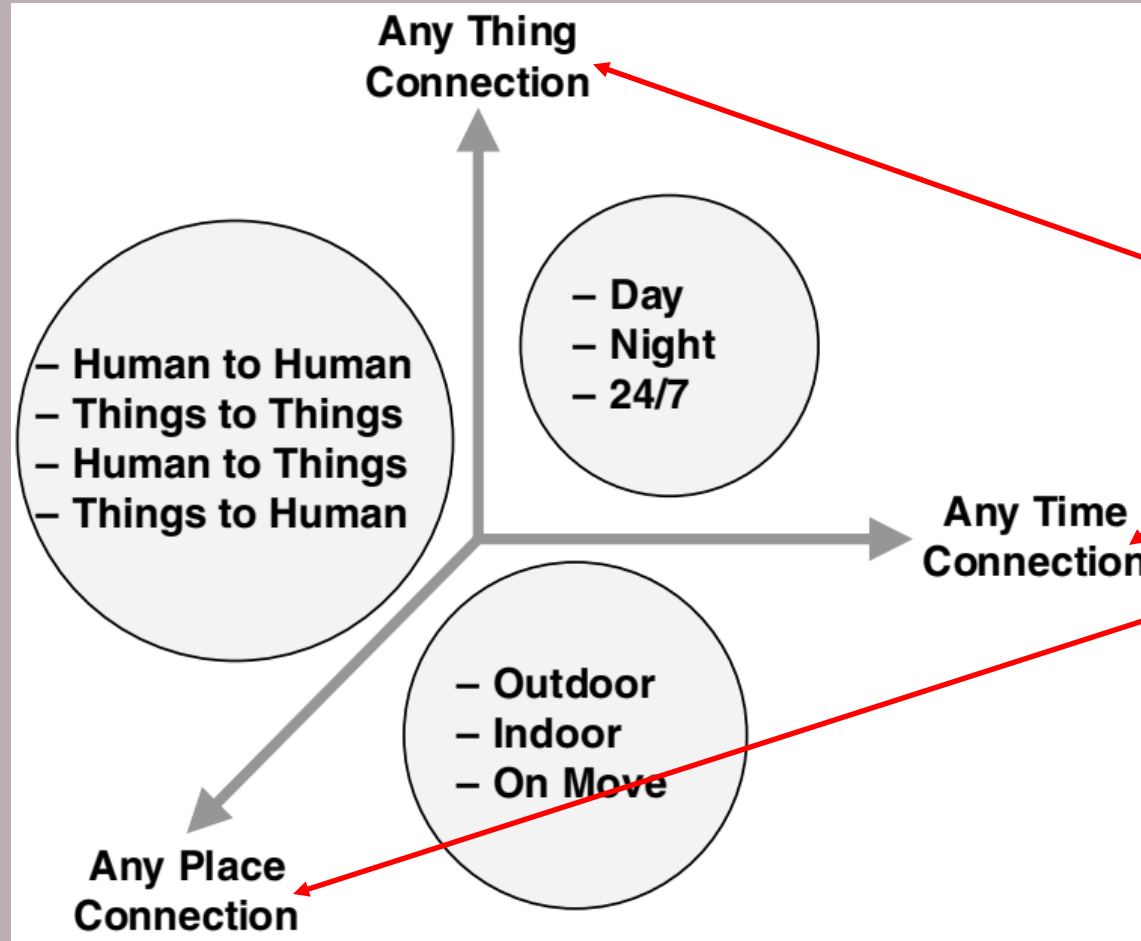


The concept of As and Cs in the IoT

Definition:

- The basic idea of IoT can be conceived as a representation of various As and Cs, as shown in the figure where
 - the A's reflect the concept of ubiquity or pervasiveness (i.e. any device, anywhere, anytime, any network etc.) and
 - the C's mirror the main characteristics of IoT (i.e. connectivity, computing, convergence, etc.).

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Definition:

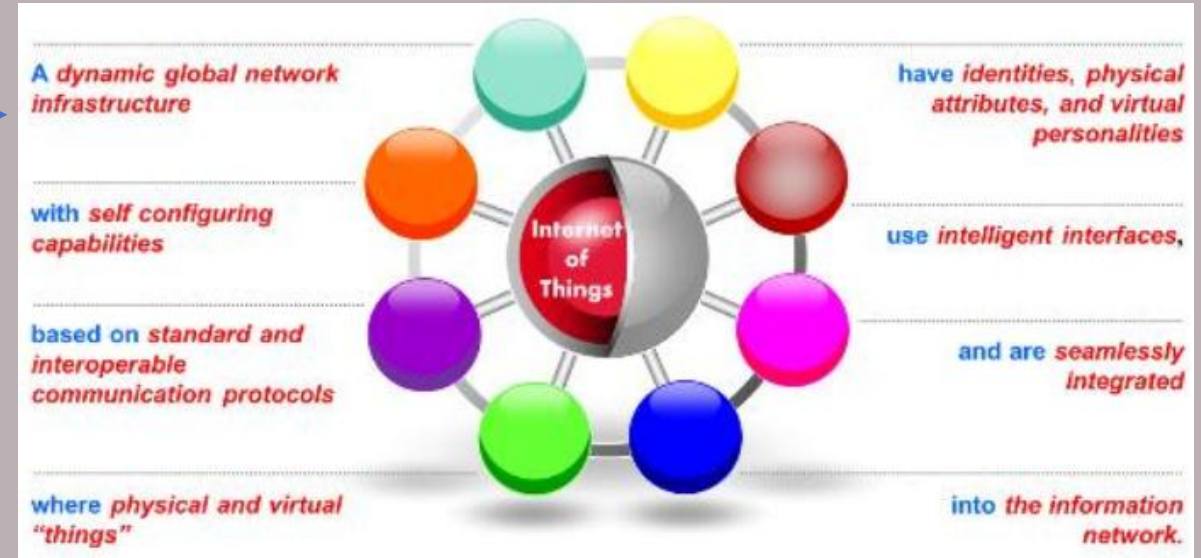
- IoT, in essence, can be seen as an addition of the third dimension named “**Thing**” to the plane of ICT world, which is fundamentally based on two dimensions of **Time** and **Place** as shown in the figure.
- This “anything” dimension ultimately boosts the ubiquity by enabling new forms of communication of humans and things and between things themselves.

Thing as a new dimension to endorse IoT

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Dynamic & Self-Adapting:

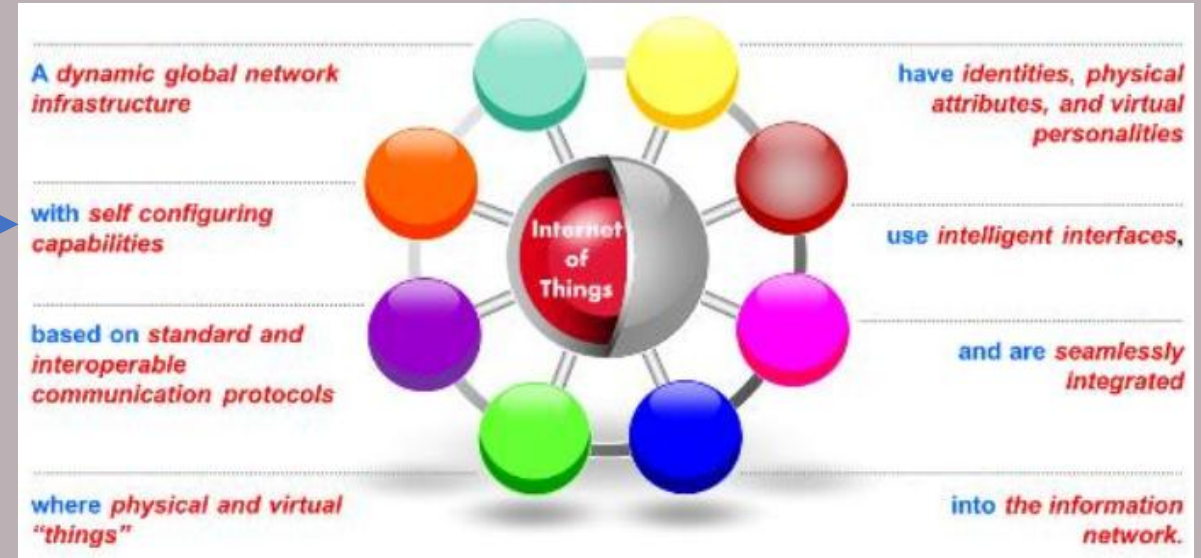
IoT devices and systems may have the capability to dynamically adapt with the changing contexts and take actions based on their operating conditions, user's context, or sensed environment.



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Self-Configuring:

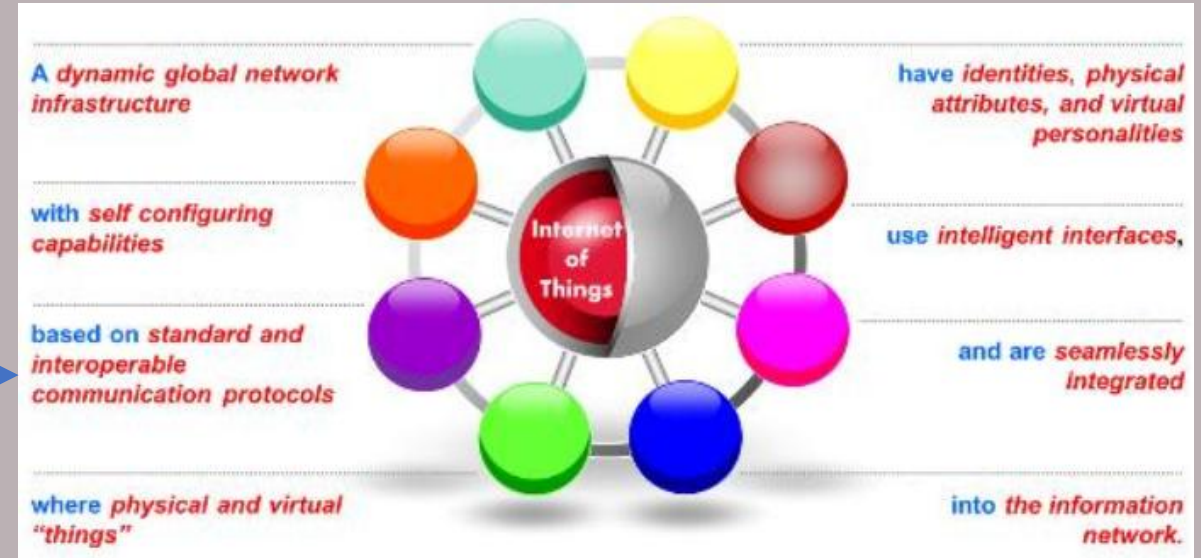
- IoT devices may have self-configuring capability, allowing a large number of devices to work together to provide certain functionality (such as weather monitoring).
- These devices have the ability to configure themselves (in association with the IoT infrastructure), setup the networking, and fetch latest software upgrades with minimal manual or user intervention.



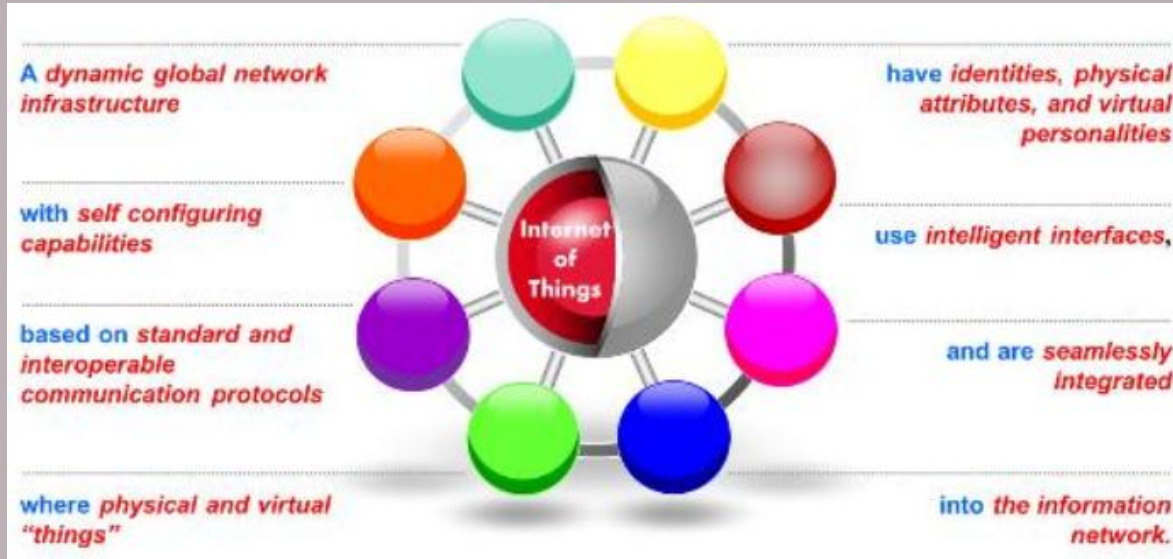
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Interoperable Communication Protocols:

IoT devices may support a number of interoperable communication protocols and can communicate with other devices and also with the infrastructure.



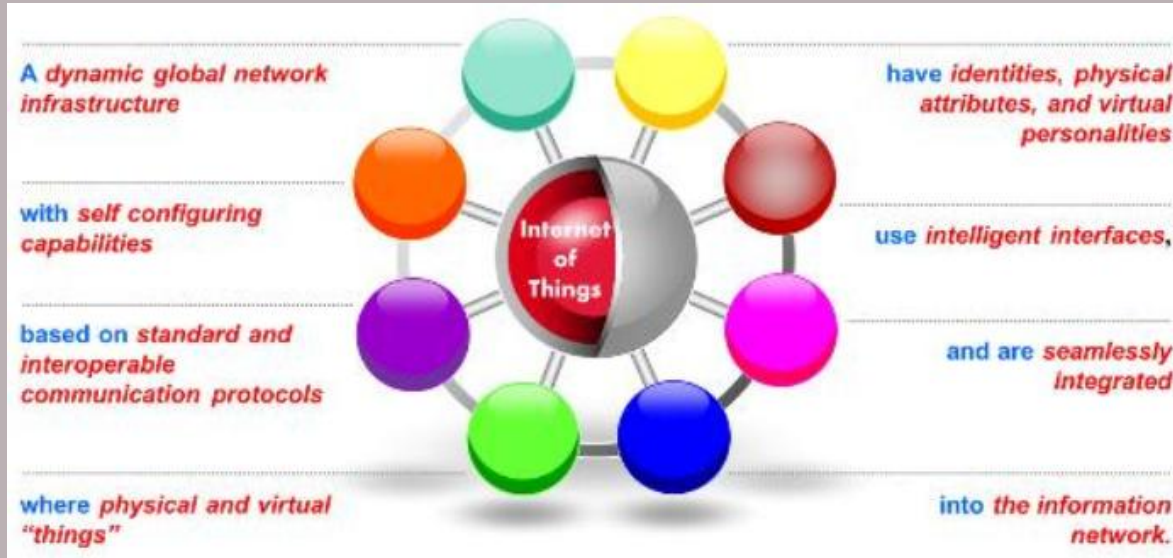
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Unique Identity:

- Each IoT device has a unique identity and a unique identifier (such as an IP address or a URI). IoT systems may have intelligent interfaces which adapt based on the context, allow communicating with users and the environmental contexts.
- IoT device interfaces allow users to query the devices, monitor their status, and control them remotely, in association with the control, configuration and management.

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Unique Identity:

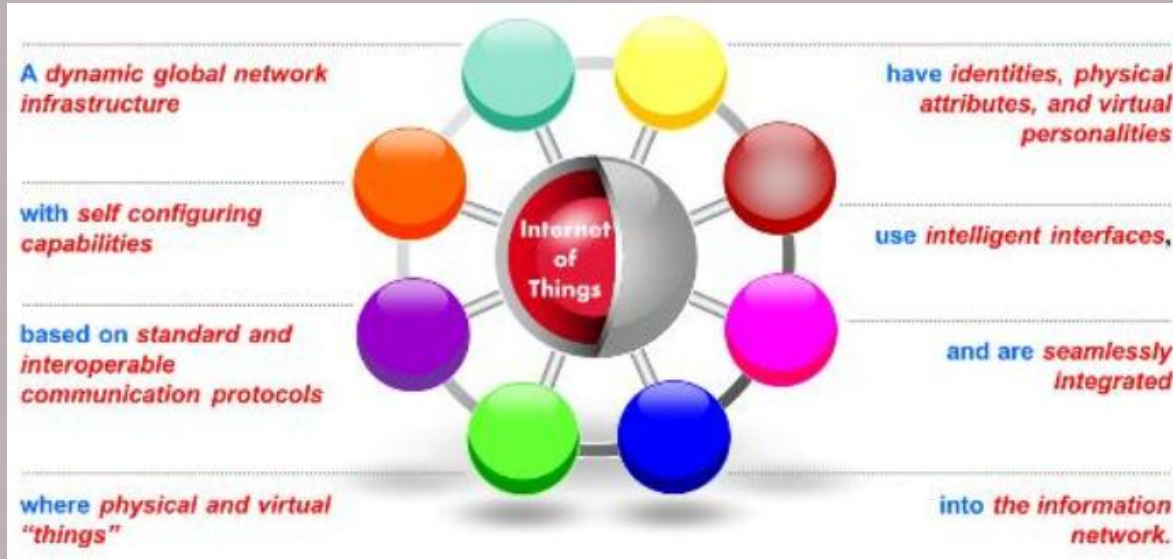
- Ubiquitous ID Centre, for instance, has implemented a 128-bit addressing system for tagging individual objects. By some calculations, the Ucode system would allow for a theoretical

340'000'000'000'000'000'000
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(340×10^{36})

addresses to be assigned. This will permit a trillion tags to be assigned every day for a trillion years, and still have some left over.

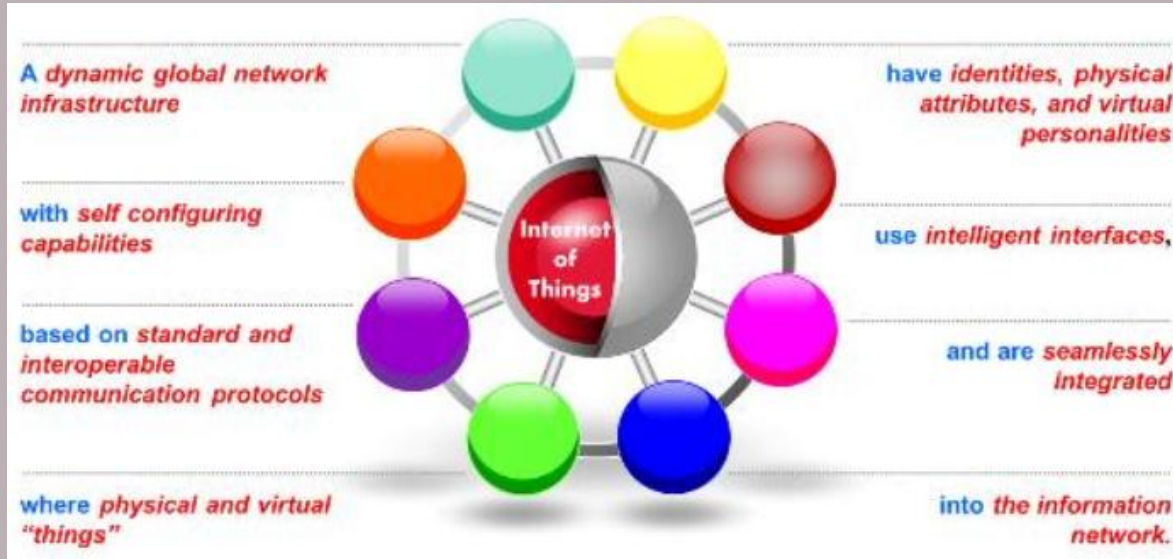
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Integrated into Information Network:

- IoT devices are usually integrated into the information network that allows them to communicate and exchange data with other devices and systems.
- IoT devices can be dynamically discovered in the network, by other devices and/or the network, and have the capability to describe themselves (and their characteristics) to other devices or user applications.
- Integration into the information network helps in making IoT systems "smarter" due to the collective intelligence of the individual devices in collaboration with the infrastructure.

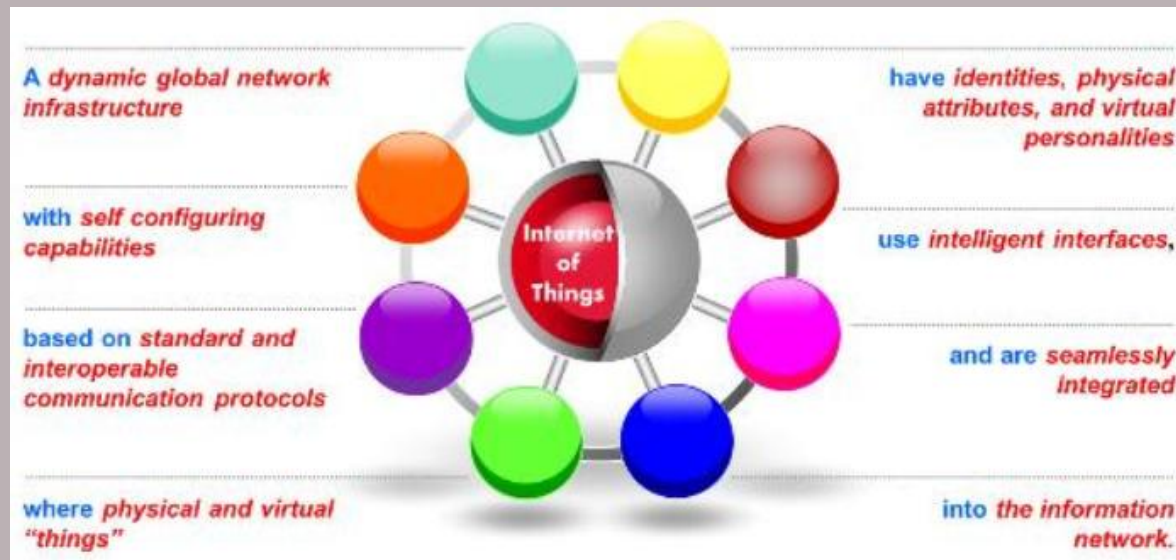
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Integrated into Information Network:

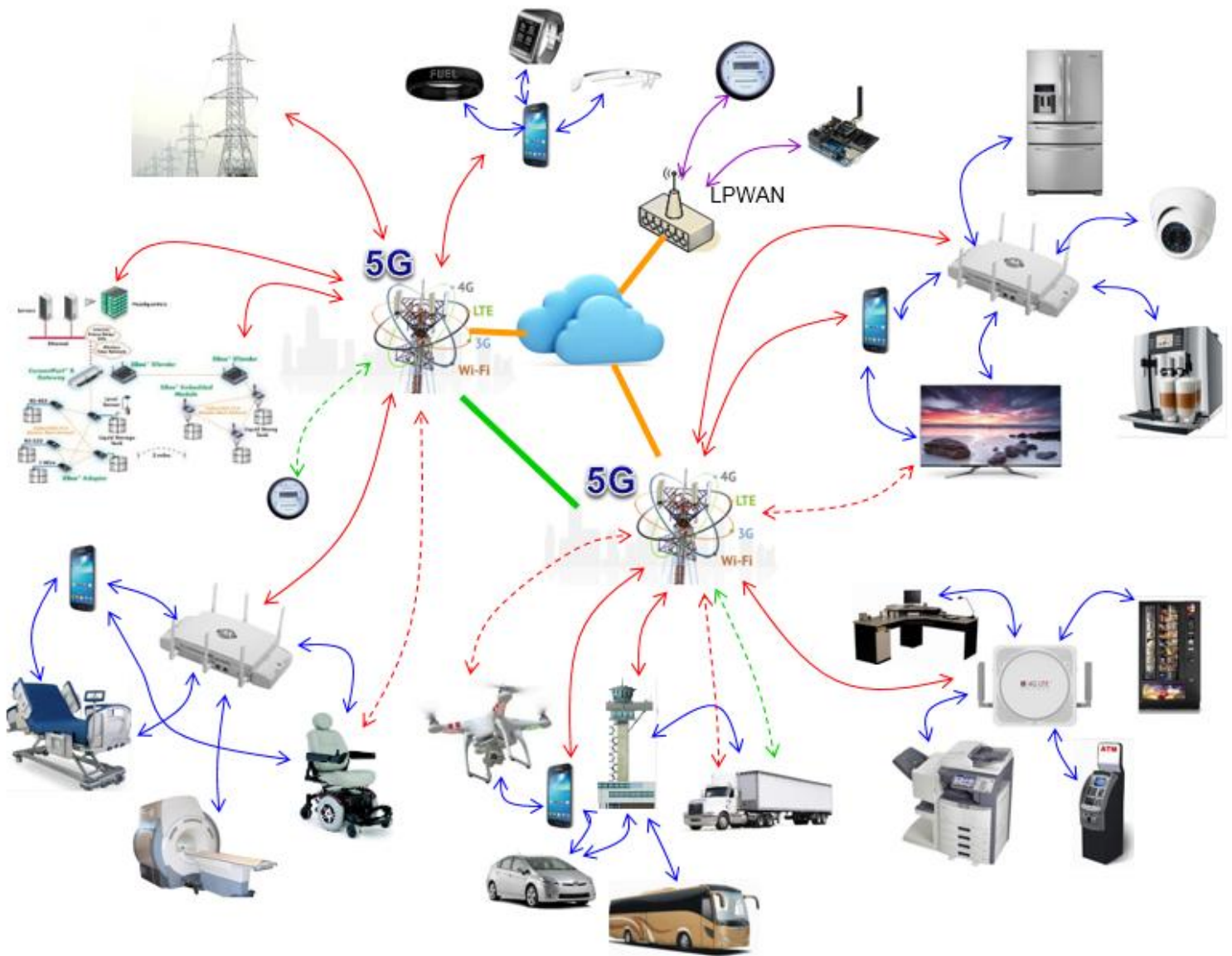


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Integrated into Information Network:





- Dotted Red** :This is cellular link (meaning communication between cellular network and the device).
- Dotted Green**: By this link, I wanted to indicated a communication in which a lot of small end device(e.g, Water meter, power meter etc)
- Solid Red**: This is also cellular link (meaning communication between cellular network and the device).
- Solid Blue**: This is a kind of short range communication being used between the device and a Gateway.
- Solid Green**: This represents a backhaul line connecting cellular tower or core network data pipe.
- Solid Violet**: This represents Non-Celluar LPWAN like LoRa, SigFox

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The IoT elements

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We can summarise the components of the Internet of Things with the following simplistic equation:

Internet of Things = Physical Object +
Controllers + Sensors +
Communication + and
Actuators + Internet

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We can summarise the components of the Internet of Things with the following simplistic equation:

$$\text{Internet of Things} = \text{Smart Object} + \text{Internet}$$
$$\begin{aligned} \text{Smart Object} = & \text{Physical Object} + \text{Controllers,} \\ & \text{Sensors} + \text{Communication,} \\ & \text{and Actuators} \end{aligned}$$